

After providing their consent, users will complete a quantitative pre-test questionnaire, then run through a Starry Story until they complete the quizzes that demonstrate their understanding of the star constellations after interacting with the learning tutor. Then, they will complete a qualitative and quantitative post-test questionnaire and an interview.

Informed Consent

Title of Study: Stars User Testing

Student Investigator: Melissa Dodson, Meiko Huang, Ruby Peeler

Faculty Supervisor: Dr. Joel Walsh

You are invited to participate in a research study conducted by Melissa Dodson, Meiko Huang, Ruby Peeler, students in the COMP 395: class at Occidental College in Los Angeles, CA. You must be at least 18 years of age to consent to participation in this study. Please read this form and ask any questions you may have before agreeing to participate in the study.

PURPOSE OF STUDY: The purpose of this study is to test our product and see what changes need to be made before making it available to the public. Our product is a learning tutor that teaches the learner about star constellations, how to identify them, and what historical background and cultural stories they have. Data collected from this study will be used to further improve our product to make it as usable, informative, and accurate as possible. We will test on a range of different users, given their familiarity with star constellations, so we have a range of data to analyze. Data derived from this study may be held for future use, and may be stored and re-analyzed, or otherwise combined with other data at a later date.

PROCEDURES: If you agree to take part in this study, you will be asked to test our product by doing a run-through of it. The whole experience will take around 30 minutes maximum and you will be asked to learn about 3 star constellations or until you complete the quizzes that demonstrate your understanding of the star constellations after interacting with the learning tutor.

VOLUNTARY PARTICIPATION: Participation in this study is voluntary. You may skip any questions that you do not want to answer or stop participating at any time. You are free to withdraw from the study at any time without penalty, with no loss of benefits to which you were otherwise entitled.

RISKS and BENEFITS: There are no anticipated risks or discomforts to your participation in this study other than those encountered in daily life.

Although you may not benefit directly from this research, by participating in this study you will help us improve our product so that we may share it with the public to improve everyone's understanding of star constellations and their historical and cultural significance and stories.

CONFIDENTIALITY: All data will be confidential unless you specifically choose to provide information attached to your identity.

COMPENSATION: You will not be paid for participating in this study.

CONTACT INFORMATION: If you have any questions or concerns about the research, you can contact Melissa Dodson at dodsonm@oxy.edu, Ruby Peeler at rpeeler@oxy.edu, or Meiko Huang at mhuang3@oxy.edu.

Pre-Test Questions

Pre-Test questions aimed to gain an understanding of the users familiarity with constellations and LLM/chatbot tools.

"How familiar are you with constellations? 1 being you know none/very few, and 5 being you could easily identify several on a given night"

"How comfortable are you using LLMs/chatbots like ChatGPT, Claude, etc.? 1 being you never/rarely use these technologies, and 5 being you use these technologies regularly (at least weekly)"

"Feel free to elaborate on your general thoughts on LLMs/chatbots."

Post-Test Questions

Post-Test Questions aimed to get feedback on the ease in which the tutor can be used, user's agency, and the knowledge they gained. We also aimed to determine if using this tool changed their opinions on LLM tutors.

"How easy/intuitive was this tool to use? 1 being you would not be able to use this tool without guidance, 5 being very easy/needed very little to no instruction."

“Do you feel as if you were directing your learning experience (i.e., did you choose the constellations and myths you engaged with, or did the tutor do that work for you?)”

“How engaged did you feel with this tool? 1 being not engaged and 5 being highly engaged”

“Do you feel as if the "tutor" facilitated an engaging learning environment? Why or why not?”

“How would you rate your knowledge BEFORE engaging with this tool? 1 being you knew little to no constellations, 5 being you recognized all the constellations mentioned.”

“How would you rate your knowledge AFTER engaging with this tool (using the same scale) ”

“Please feel free to elaborate more on the knowledge you gained. If there is anything that you feel would have better supported your learning experience, please let us know!”

“Please feel free to elaborate on the features you found beneficial/hindering to your learning (e.g., the interactive sky, the journal, etc.)”

“Was there anything that stood out to you? Anything that should be changed? What should be kept?”

“Is there anything that we could add to this tool to make it better in any way?”

“Did this experience change the way you view learning with LLMs? Why or why not?”

Anonymized User Testing Notes are in User Testing Post-Test (Responses) - Form Responses 1.csv